# Machine Learning Project

*Predicting the GDP Evolution*

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# Agenda

- Problem Statement – Project Details
- Getting to know the data
- Forecasting
- Limitations and Improvements

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**Problem Statement – Project Details**

Getting to know the data

Forecasting

Limitations and Improvements

# Project Details

From where did we get the data and what did it include?

- GDP  and Population data downloaded from World Bank website
- 60+ years of history starting from 1960
- Non-country entities in data (European Union)
- 250+ countries, regions and economic blocs examined
- 13,500+ data points/observations used

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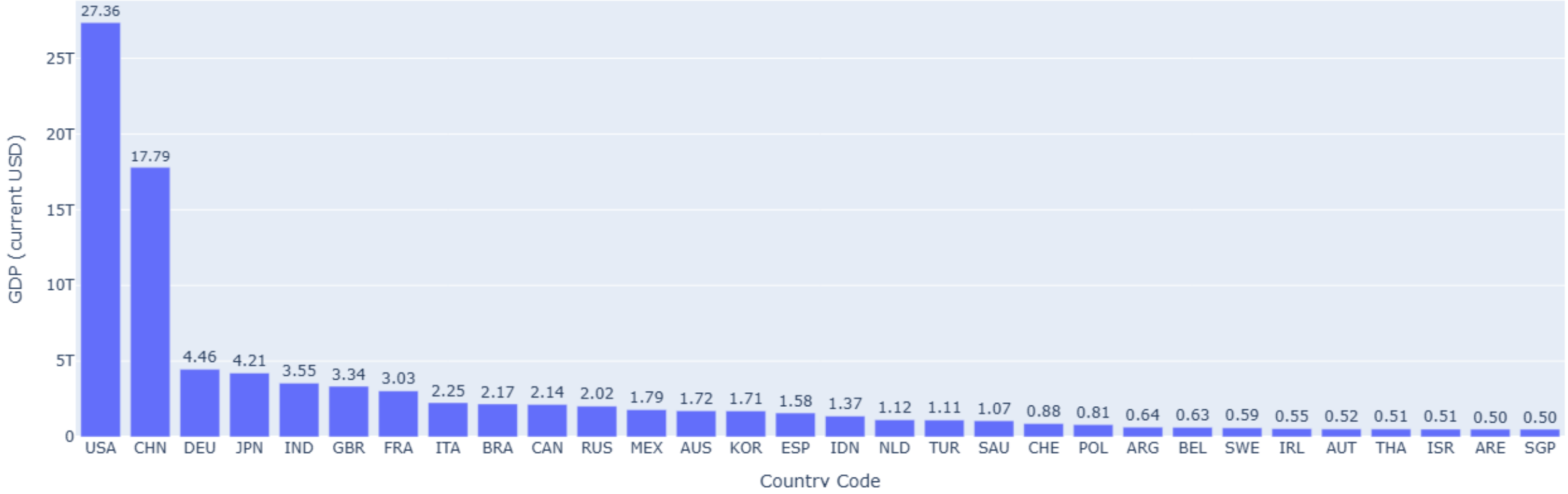
# Getting to know the data: GDP per country

US and China the countries with highest GDP by far; >3 times as high when compared to 3rd Germany

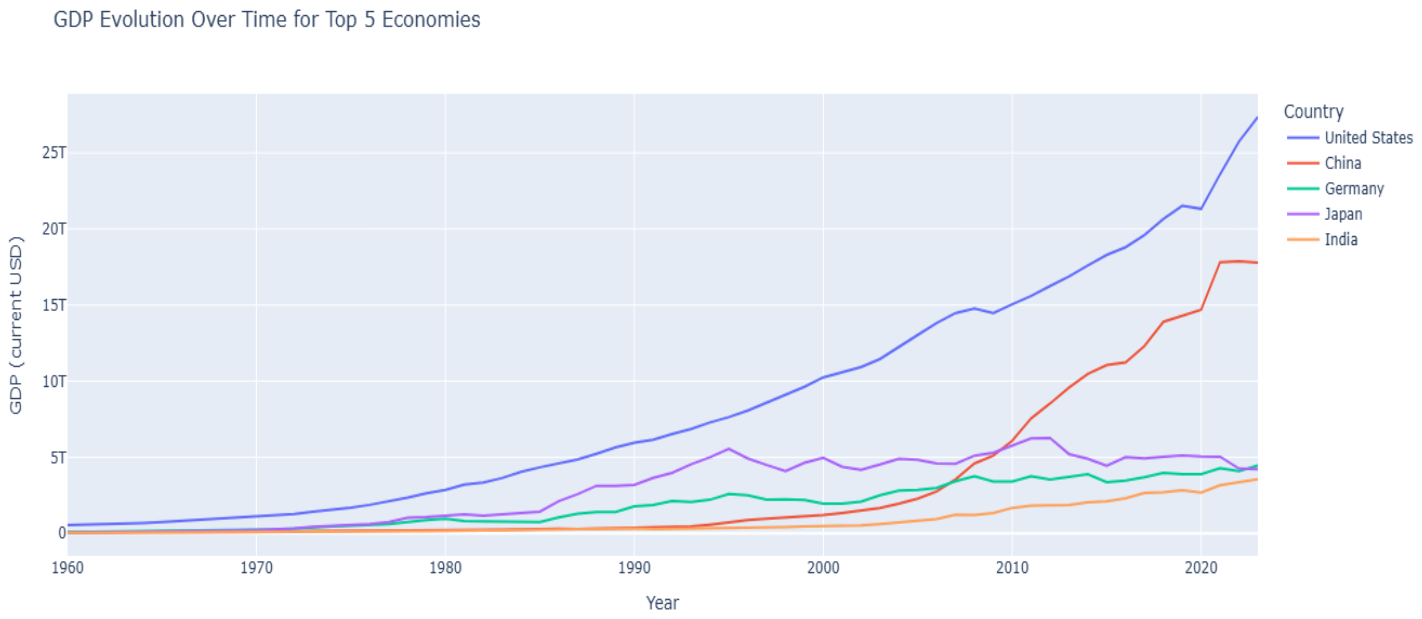
**Gross Domestic Product (in Trillion USD)**

*Top-30 Countries, 2023*



# Getting to know the data: GDP Growth per major Country

US and China with highest overall growth over the past 60 years albeit with flattening trends recently

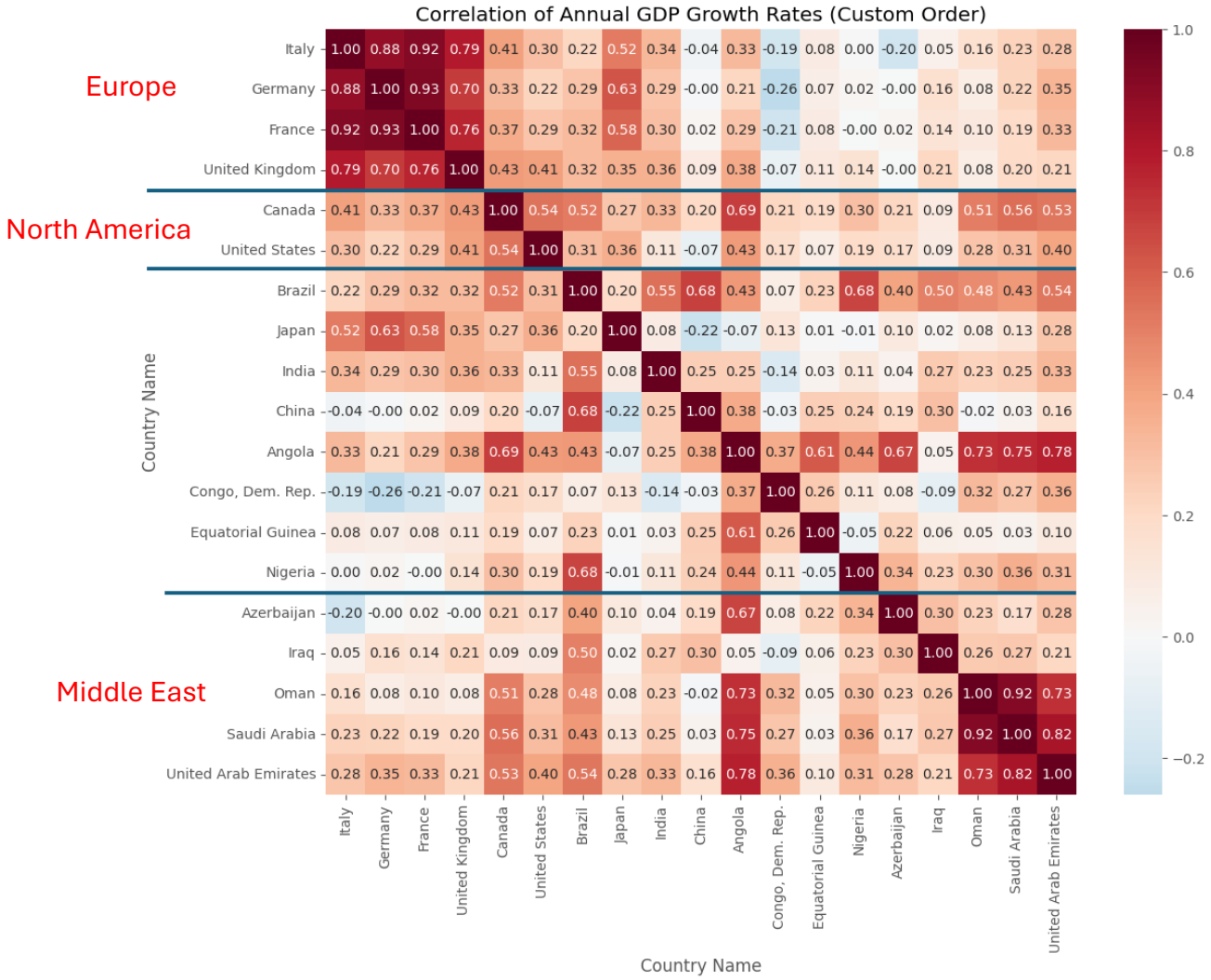



- US steady growth with a CAGR of 6.4%
- China's growth "exploded" since 2000 although with major slow-down in the past two years
- Japan GDP, although with high growth until 1990, remains almost flat since then
- Germany with small but steady growth and a CAGR of 6.5%
- India GDP grew exponentially since 2000 at a CAGR of 7.51%

| CAGR (1960-2023) | USA | China | Germany | Japan | India |
|---|---|---|---|---|---|
| | 6.42% | 9.46% | 6.50% | 7.38% | 7.51% |

# Getting to know the data: GDP Growth correlation matrix

## GDP growth within certain country groups highly correlated




- European countries' GDP growth very strongly correlated
- Middle East countries with high growth correlation
- China growth patterns not correlated with other major economies but Brazil
- India and Japan growth more correlated with EU countries vs. US

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# Four forecasting models used and three geographic entities chosen

| Models used | Countries tested | Timeframe |
| --- | --- | --- |
| • Linear Regression<br>• Polynomial Regression<br>• Autoregression Integrated Moving Average (ARIMA)<br>• Prophet | • United States<br>• European Union<br>• Switzerland | • Training period for 50 years (since 1960)<br>• Testing period from 2010 until 2023<br>• Five-year forecasting period up to 2028 |

# What does each model do?

## Linear Regression

- Calculates the line of best fit by minimizing the vertical distance
- Predicts based on assumption that trends will continue linearly
- Relationship between data is linear

## Polynomial Regression

- Captures non-linear relationships
- Makes predictions by raising existing variables to higher powers

## ARIMA

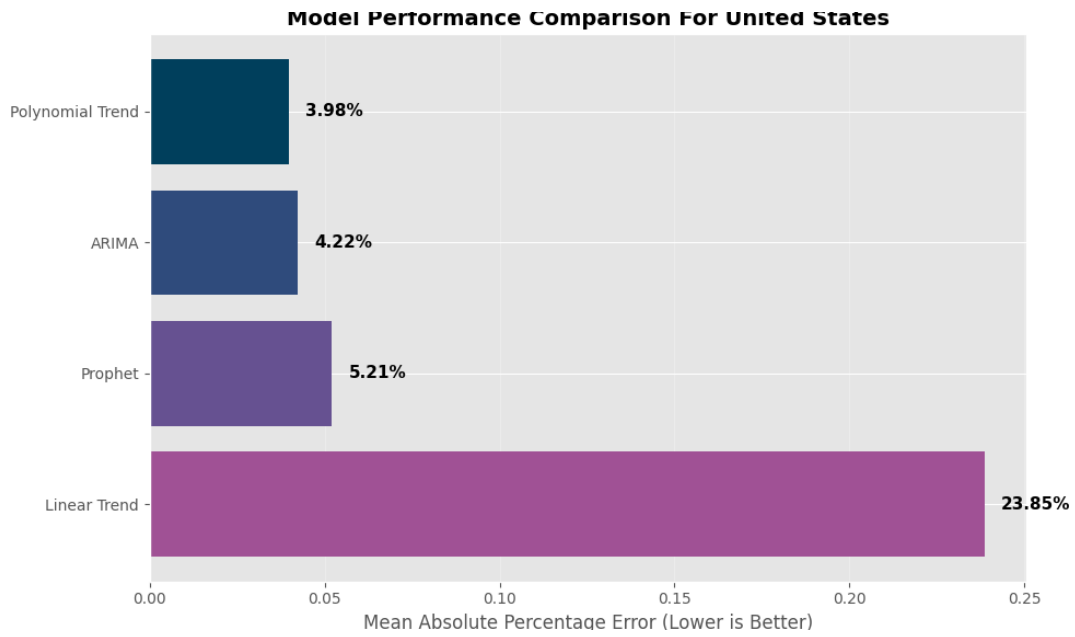
- Uses previous values to predict next values
- Takes differences to make the data stationary
- Uses past forecasts to improve predictions

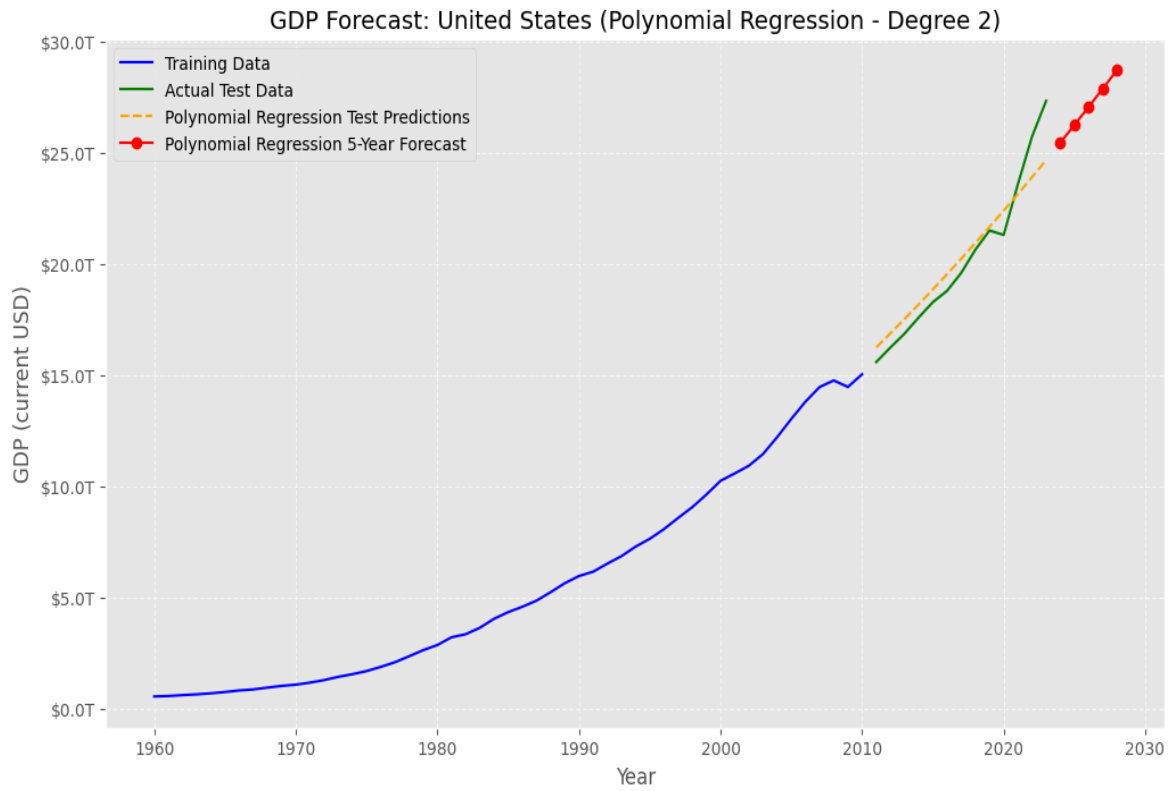
## Prophet

- Combines trend, seasonality and holiday effects
- Adjusts for holidays and specific one-time spikes
- Detects patterns and makes predictions accordingly

# Forecasting (I): US

Polynomial Regression with lowest predictive error capturing the US GDP growth trend



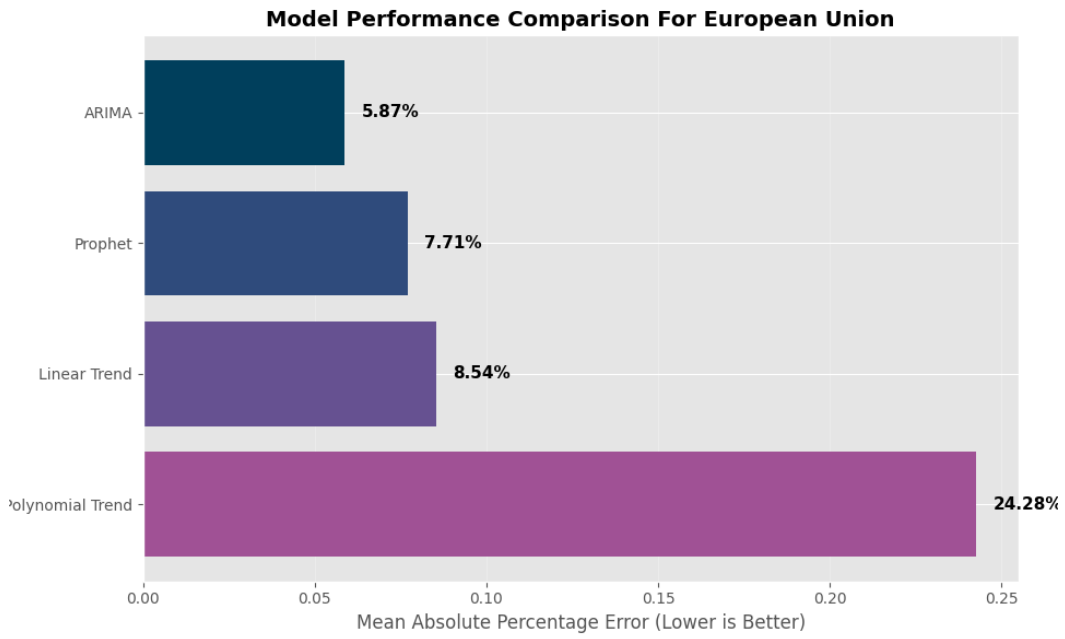



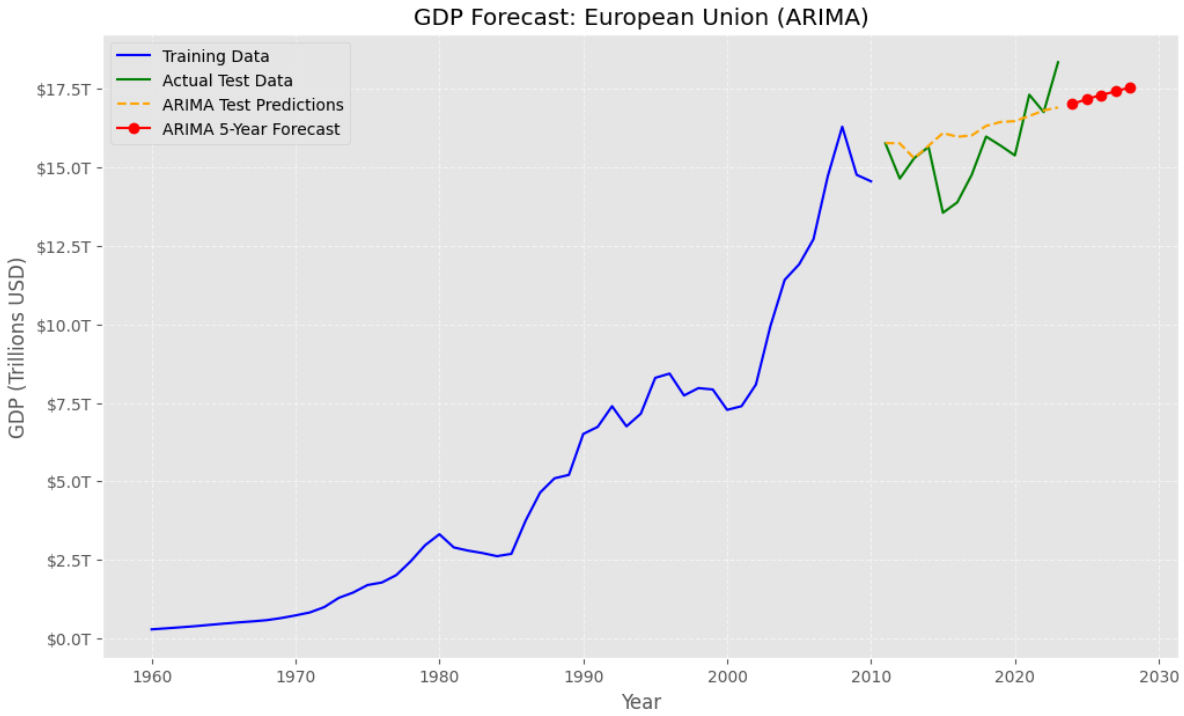

## Key findings

- Polynomial regression the best fit among the models tested
- Linear Regression has a large error since the US's GDP growth is exponential
- ARIMA and Prophet more stable models

# Forecasting (II): European Union

ARIMA with the lowest predictive error (6%), but not fully capturing future growth

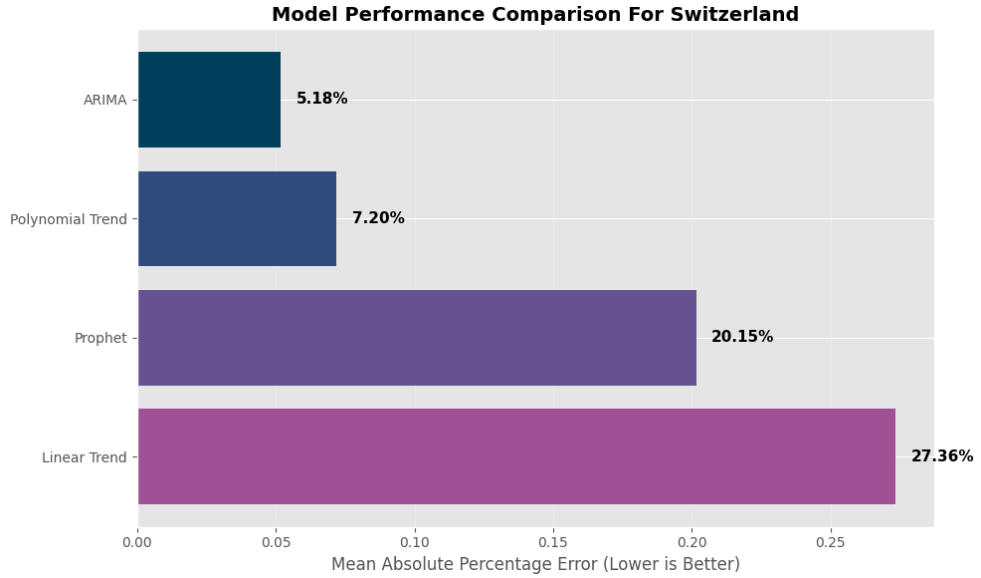


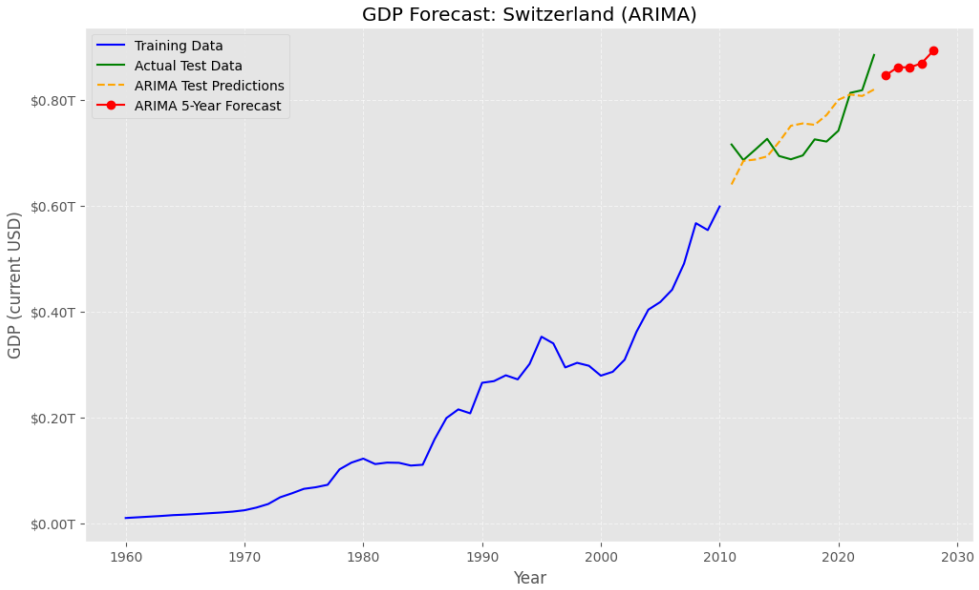




- European Union's trend seems to be linear
- Most models performed well and had similar results apart from Polynomial Regression with an error of ~25%
- ARIMA predicts well the growth trend of the EU GDP

# Forecasting (III): Switzerland

ARIMA with the lowest predictive error (5%), while linear regression not as accurate (27% error)




- Switzerland's growth is somewhat exponential
- A linear trend does not have good results, but a polynomial does
- ARIMA very good predictions and close to actual results

# Summarizing Model Accuracy

ARIMA with lowest predictive error across countries (90% predictive accuracy)

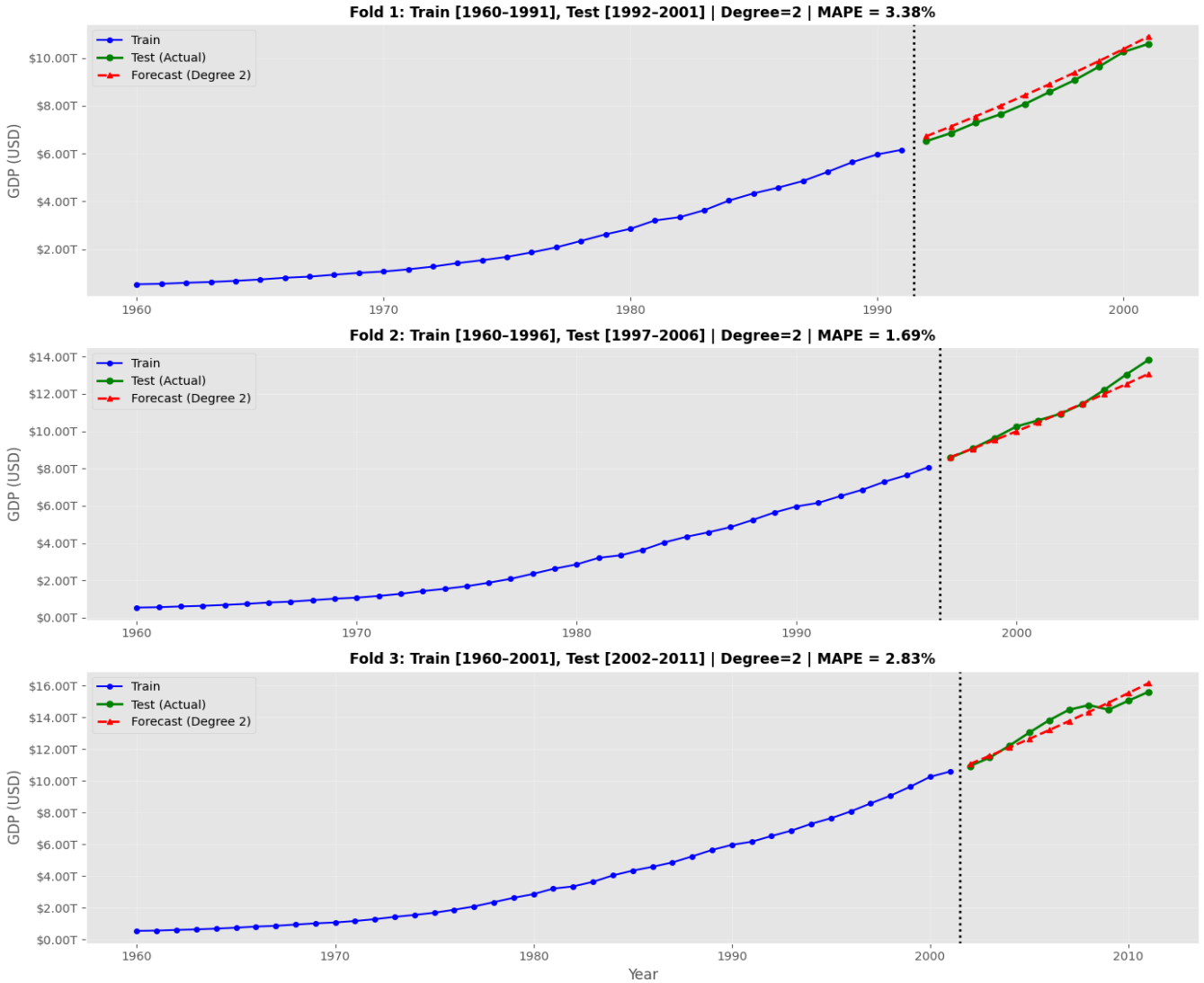
| Country | Linear MAPE (%) | Polynomial MAPE (%) | ARIMA MAPE (%) | Prophet MAPE (%) |
|---|---|---|---|---|
| Italy | 12.36 | 41.05 | 5.55 | 20.46 |
| Germany | 6.56 | 15.72 | 4.6 | 5.21 |
| France | 6.73 | 26.75 | 5.14 | 11.41 |
| United Kingdom | 6.33 | 34.21 | 4.94 | 15.72 |
| Canada | 24.65 | 13.34 | 8.43 | 17.62 |
| United States | 23.85 | 3.98 | 4.22 | 5.21 |
| Brazil | 37.39 | 73.58 | 9.53 | 37.67 |
| Japan | 34.48 | 34.69 | 7.08 | 34.54 |
| India | 59.07 | 32.99 | 5.45 | 58.86 |
| China | 75.48 | 22.62 | 4.01 | 75.47 |
| Angola | 88.52 | 112.83 | 30.99 | 88.44 |
| Congo, Dem. Rep. | 28.23 | 6.47 | 6.51 | 28.24 |
| Equatorial Guinea | 27.69 | 216.12 | 18.09 | 27.37 |
| Nigeria | 44.99 | 22.57 | 12.44 | 44.85 |
| Azerbaijan | 37.25 | 89.94 | 27.17 | 37.19 |
| Iraq | 60.42 | 53.06 | 15.89 | 60.37 |
| Oman | 46.84 | 14.52 | 10.49 | 46.75 |
| Saudi Arabia | 52.14 | 31.09 | 10.93 | 52.06 |
| United Arab Emirates | 32.07 | 65.64 | 8.09 | 32.01 |
| Average | 37.11 | 49.96 | 10.50 | 36.81 |

- ARIMA achieved the lowest average error and remained the most stable across all countries
- Polynomial Regression was the worst performer overall
- Forecasting was accurate for Western countries but faced significant challenges in African markets

# Backtesting

## Backtesting data show that Polynomial Regression is a stable model for US GDP prediction

- Best Model: Polynomial Regression for the US
- Backtesting
- Train size increases with every fold
- Test size remains the same
- Shows stability of model



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# Limitations and Improvements

## Challenges Identified and Opportunities for Enhancement

### Limitations

- **Ignores the "Real world"**
  - Wars
  - Elections
- **Vulnerability to "Black Swan" events**
  - COVID – 19
  - 2008 economic crisis

### Improvements

- **Add more advanced models**
  - LSTM
  - Prophet with use of MCMC
  - Other neural networks
- **Add more data to improve the models' predictions**
  - Population
  - Unemployment
  - Inflation

# THANK YOU !